	Solving the equations simultaneously, $r = 3.0\ \Omega$ and $E = 6.0$ V
19	**A** The equivalent capacitance of the right branch is $(\frac{1}{C}+\frac{1}{C})^{-1}+C=\frac{3C}{2}=1.5C$ Since charge is the same at the right and left branch, $Q_{right} = Q_{left} = 6\ \mu C$ $V_{right}=\frac{Q_{right}}{C_{right}}=\frac{6}{1.5C}=\frac{4}{C}$ $Q_{bottom}=\frac{C}{2}V_{right}=\frac{C}{2}(\frac{4}{C})=2\mu C$ Since the two capacitors in the bottom branch are in series, each capacitor stores the same amount of charge, capacitor Y has 2 μC
20	**A**